

41026 Advanced Software Development

Prediction Market Intelligence Hub

Release 1: MCP, RAG & Intelligent Agent Integration

Assessment 2: Release 1 (30%)

Technical Report

Team Name: Group 29

Workshop Class / Tutor: Grace Billiris

Student #	Full Name	Student ID	GitHub Username	Assigned Feature
Student 1	Alex Choi	26255859	AlexChoi2004	Market Watchlist & Discovery
Student 2	Christopher Kumarelil Cherian	25491198	christopherkumarelilcherian-bit	Portfolio & Position Tracker
Student 3	Syed Humza Hussain	25474457	SHumzaHussain – Silv3r-exe	Market Research Notes & News Feed
Student 4	Erfanur Rahman	24951459	Erfanur1	AI Market Analyst Assistant

GitHub Repository URL: <https://github.com/Erfanur1/ADVSD>

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1. Project Overview and Release 1 Scope

1.1 Release 0 Recap

Prediction Market Intelligence Hub is an Agentic AI microservices application for tracking, researching, and analysing prediction markets. Release 0 established four containerised student features (frontend, backend/API, and an embedded SQLite database each), a shared local AI-Mode service (Ollama), and the Plan-Act-Observe-Adapt agentic loop, orchestrated locally via Docker Compose.

1.2 Release 1 Scope

Release 1 extends the Release 0 application with two new shared, non-containerised, local services: an MCP (Model Context Protocol) server exposing a set of registered tools, and a RAG (Retrieval-Augmented Generation) server that grounds AI answers in retrieved project context with source citations and a confidence category. The existing Release 0 shared agentic loop is extended with dedicated MCP and RAG validation modes. Each student frontend gains the ability to invoke MCP tools and RAG queries through its own backend/API; the underlying containerized feature microservices from Release 0 remain unchanged and operational.

As part of this release, the team moved from seeded/mock market data to live market data from two public, no-authentication APIs: Polymarket and Kalshi. A shared data layer (`ai-services/shared/market_data.py`) fetches both, converts them into one common market format (title, probability, bid/ask, volume, close date) and caches results for 5 minutes to protect against rate limits and downtime. Manifold was used at first, but Student 4 replaced it with Kalshi, a regulated real-money exchange. Kalshi gives real order-book prices and allows cross-exchange comparisons with Polymarket. Its public market data endpoints need no account.

1.3 Feature Updates and Responsibilities

Student	Individual Release 1 Update	Shared Group Responsibility
Student 1	Integrated MCP and RAG into Market Watchlist & Discovery. Added live market search (MCP <code>search_markets</code> , Polymarket + Kalshi data) and grounded Q&A (RAG, with citations, confidence, and insufficient-context fallback). Verified end-to-end via frontend, terminal, and Docker Compose. Regression-tested Release 0 CRUD and AI-Mode — no breakage. Ran the agentic loop in both MCP and RAG modes. Updated <code>student-1.yml</code> and <code>docker-compose.yml</code> with MCP/RAG config.	Built and own all three shared components: MCP server (tool registry), RAG server (retrieval, grounding, citations, confidence), and the extended agentic loop (MCP + RAG validation modes). Also built the underlying live data layer (Polymarket + Manifold originally, later updated from Manifold to Kalshi by Student 4).
Student 2	Upgraded the UI with HTMX-driven panels for "Market Intelligence" (RAG integration for historical asset queries) and "Portfolio Validation" (MCP tool execution). Added a dynamic Chart.js profit/loss graph for position management. Configured local networking (<code>host.docker.internal</code>) to route backend requests to the shared servers.	Added MCP tools <code>validate_portfolio</code> and <code>get_exposure_summary</code> ; added the RAG portfolio loader
Student 3	Extended Market Research notes and News feed with frontend-to-backend-to-shared service access to both the MCP server and RAG server.	Research notes/news knowledge-base loader, <code>load_notes_entries</code>

	Added a "Live markets (MCP)" search panel that calls a new POST /mcp/search backend route which in turn calls the shared MCP servers search markets tool and returns live market results. Added an "Ask, grounded in your notes RAG" panel that calls a new POST /rag/ask backend route which calls the shared RAG server /rag/query endpoint and displays a grounded answer with citations and a confidence score , and or a visually distinct insufficient context message when no relevant context is found. Also contributed the load-notes-entries() loader in the shared RAG server (ai-services/rag-server/app.py which pulls this features own research notes and news articles live from its /notes and /news endpoints and folds them into the shared knowledge base used by every features RAG queries – supplying the notes/news half of the RAG servers grounding data for the whole team.	
Student 4	Added cross-exchange mispricing detection to the AI Market Analyst Assistant. A "Cross-exchange mispricing" panel searches Polymarket and Kalshi through MCP (search_markets). The user picks the same event on each exchange and compares them with the new MCP tool compare_markets. The panel shows the verdict (mispriced / watch / fair), the price gap, the edge left after the spread, and both bid/ask books. A comparison can be saved as an analysis. Its "Explain this gap" button sends both markets and the measured gap to the RAG server, which returns a grounded explanation that cites both sides. An "Ask with sources" panel gives grounded RAG answers with citations, a confidence label and a distinct insufficient-context message. Every AI, MCP and RAG call shows its Plan → Act → Observe → Adapt trace. The backend adds POST /mcp/execute (limited to search_markets, get_market_price and compare_markets), POST /mispricing/save and POST /rag/ask. Each returns a graceful fallback if a shared server is unreachable. All database and service output is now HTML-escaped, which closes a script-injection bug in the Release 0 page.	Replaced Manifold with Kalshi in the shared live data layer (ai-services/shared/market_data.py). Kalshi has no search endpoint, so the data layer builds a background index of every open Kalshi event (refreshed every 30 minutes, and kept as-is if Kalshi is down). Also fixed the Polymarket search, which was returning unrelated markets, and added the offline data-layer test suite. Added the compare_markets MCP tool and its tests. Extended the RAG server to accept up to three market_ids, so both sides of a comparison are retrieved and cited. Led the shared UI refresh (one theme, shared sidebar, dashboard home) across all four frontends (PR #16).

2. Release 1 Requirements

2.1 Shared MCP Server

- The system shall run one shared, non-containerised, local MCP server, used by all four student features.
- The MCP server shall expose a defined set of registered tools with enforced boundaries (i.e. tools only perform their documented action, nothing else).

- The MCP server shall accept valid tool-call requests and return structured results (not free text).

2.2 Shared RAG Server

- The system shall run one shared, non-containerised, local RAG server, used by all four student features.
- The RAG server shall retrieve relevant project context (e.g. research notes, news snippets, market descriptions) before generating a response.
- Every RAG answer shall include source citations and a confidence category (e.g. High / Medium / Low).
- If no relevant context is found, the system shall return an explicit insufficient-context response rather than an unsupported generated answer.

2.3 Frontend and Backend/API Integration

- Each feature's frontend shall provide a way to trigger at least one MCP tool call and at least one RAG query.
- Each feature's backend/API shall forward MCP and RAG requests to the shared local servers and return the result to the frontend.

2.4 Shared Agentic Loop

- The Release 0 agentic loop shall be extended with a dedicated MCP validation mode that exercises at least one MCP tool call end-to-end and logs the Plan-Act-Observe-Adapt stages.
- The Release 0 agentic loop shall be extended with a dedicated RAG validation mode that exercises at least one RAG query end-to-end and logs the Plan-Act-Observe-Adapt stages.
- The agentic loop shall run locally and shall not be containerised.

3. Non-Functional Requirements

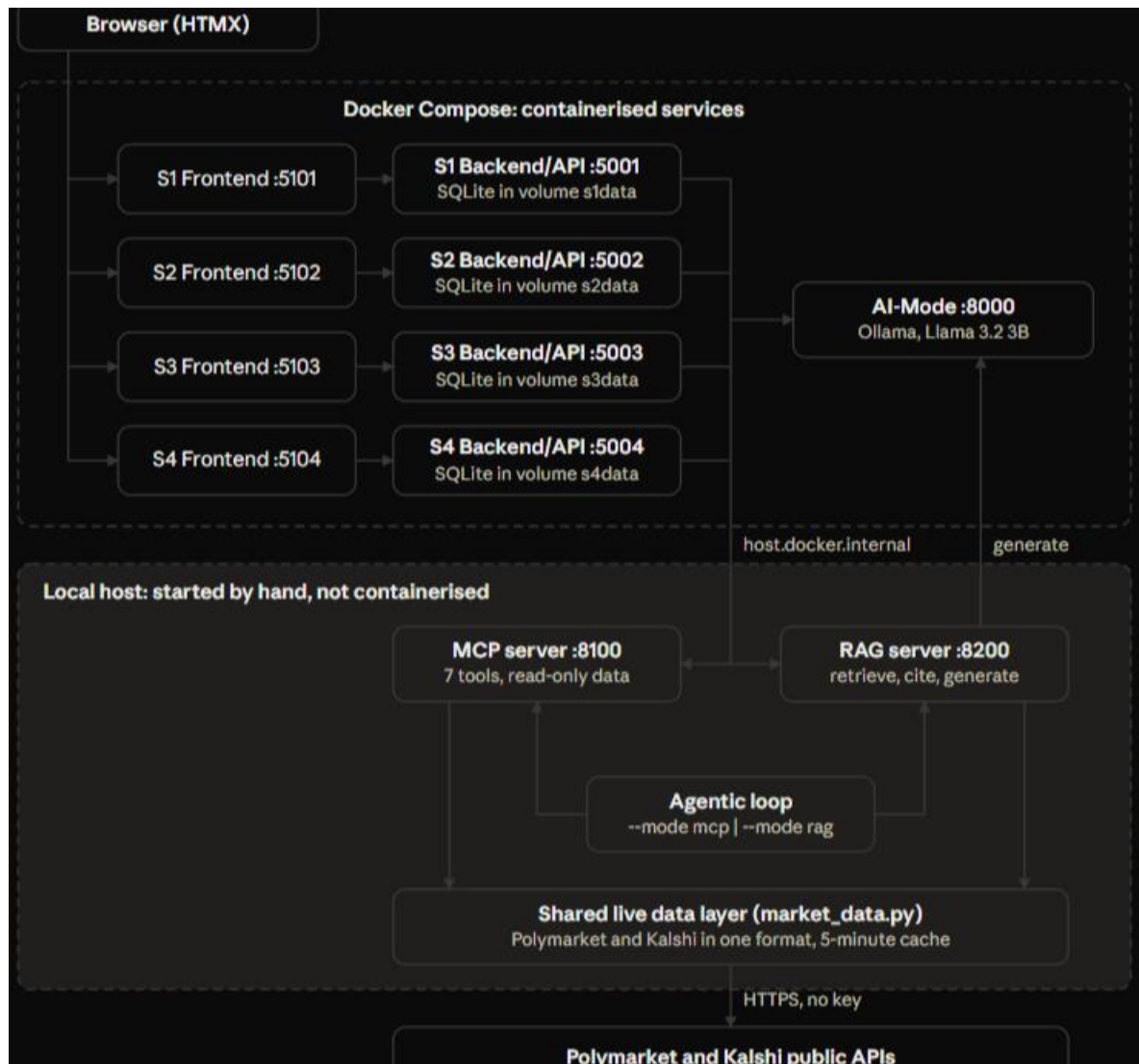
Category	Requirement	Justification
Security	MCP tools shall only access data explicitly required for their documented function (e.g. a market-price tool shall not also expose other users' portfolio data). No credentials are stored: Polymarket and Kalshi are read through public endpoints. The servers bind to 0.0.0.0 so containers can reach them, so they run only on a trusted local network. LLM output is HTML-escaped before display (0 unescaped fields in the Student 2 frontend).	Avoids storing secrets and prevents script injection through model or user text.
MCP Tool Boundaries	Each registered MCP tool shall have a clearly documented input schema and output schema; the MCP server shall reject malformed or out-of-scope requests.	Limits what any feature or agent can do through MCP to documented actions.
RAG Grounding & Traceability	Every grounded RAG response shall be traceable to at least one cited source in the knowledge base; ungrounded responses shall not be presented as factual.	An answer can always be traced to the data that produced it.
Reliability	If the MCP or RAG server is unreachable, the calling backend/API shall return a graceful fallback message rather than an unhandled error (consistent with the Release 0 AI-Mode fallback pattern).	Matches the Release 0 AI-Mode fallback pattern; demos survive outages.
Performance	MCP tool calls shall return a result within 5 seconds under normal local conditions; RAG queries shall return within 15 seconds given local LLM inference time.	Local CPU in inference is the bottleneck; the cache

		avoids repeated external calls.
Live Data Reliability	If the Polymarket or Kalshi API is unreachable, rate-limited, or slow, the system shall serve the most recently cached market data rather than failing the request.	Will still provide recent data instead of leaving users stranded and also gives a sense of when connection was lost.
Usability	MCP tool results and RAG grounded answers shall be presented in the frontend in a clearly distinguishable format from plain AI-Mode chat responses (e.g. showing citations and confidence visibly).	Users can tell a grounded answer from a plain AI-mode reply
Maintainability	The MCP server's tool registry shall be defined in a single, clearly structured location, so new tools can be added without modifying calling code.	Each student extends shared services in a marked section without merging conflicts.
Interoperability	One contract for all features: MCP {tool, params} → {result} or {error}; RAG {question} → {answer, citations, confidence, insufficient_context}. Live markets share one format for both exchanges.	Any backend or agentic loop, can call any tool or query.
Availability	The MCP and RAG servers shall be started independently of Docker Compose and documented with clear startup instructions so any team member can run them locally.	Any team member can run the stack; CI never depends on local server

4. Release 1 Architecture

The Release 1 architecture extends the Release 0 containerised application with two new shared local services (MCP and RAG) plus the extended agentic loop, all running outside Docker Compose on the local host. The existing containerised student microservices connect to these local services over HTTP, using the same connection pattern already established for AI-Mode in Release 0.

By decoupling the intelligence servers (RAG/MCP) from the containerized application stack, the architecture allows the heavy local inference workloads (Ollama) and live-data caching mechanisms (Polymarket/Kalshi) to run natively on the host machine's hardware. The frontends remain lightweight, utilizing HTMX to dispatch requests to their respective Flask backends, which package the payloads and forward them to the host-level shared servers.



The top dark zone is the docker compose; the shaded bottom zone runs on host. Browser to frontend is HTML; every arrow behind the frontend carries JSON over http, and the backend reaches the host through `host.docker.internal`. Each backend keeps its own SQLite volume. The MCP and RAG servers read feature data back through published ports, so no service touches another features database. The MCP server, RAG server, and agentic loop are not defined in `docker-compose.yml`. AI-mode and Ollama are still defined in there from release 0; section 9 records this.

4.1 Updated Repository Structure

The repository structure extends the Release 0 layout with new top-level folders for the shared MCP and RAG servers and the extended agentic loop, none of which are referenced in `docker-compose.yml`:

```

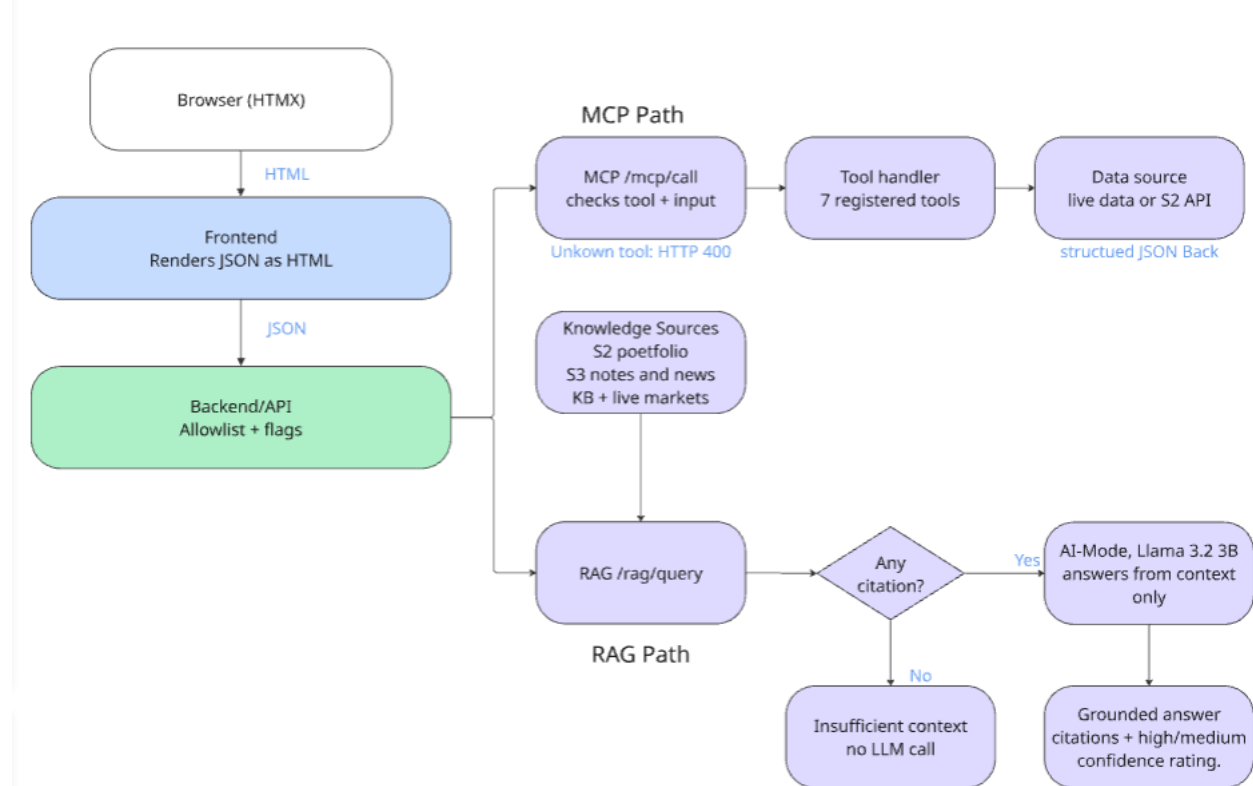
prediction-market-hub/
+-- ai-services/
|   +-- ai-mode/           # Release 0, unchanged
|   +-- mcp-server/        # NEW: shared local MCP server
|   +-- rag-server/        # NEW: shared local RAG server
|   +-- agentic-loop/      # extended with MCP + RAG validation modes
+-- student-1/ ... student-4/ # Release 0 containerised services, unchanged structure
+-- docker-compose.yml      # updated: backend env vars for MCP_URL / RAG_URL added

```

+-- docs/reports/release-1/

5. MCP and RAG Design

Both flows follow one rule: the browser talks HTML to its frontend, and everything else behind communicates via JSON, with the backend/API as the only gateway to the shared servers.



Both paths start at the same backend. MCP returns a structured tool result with no LLM involved; RAG calls the LLM only after retrieval has found at least one citation. Both results return to the backend as JSON, and the frontend renders them as HTML.

5.1 Registered MCP Tools (proposed starting set)

Tool	Input	Output	Owner
get_market_price	market_id, source	Current probability, volume, and category for the given market, sourced live from Polymarket or Kalshi (with a short local cache).	Student 1 · S1, S3, S4, loop
search_markets	query, category (optional), source (optional)	List of matching markets with id, title, probability, sourced live from Polymarket and/or Kalshi.	Student 1 · S4
get_market_history	market_id, days, source	Historical probability points for the given market,	Student 1

		sourced live from Polymarket or Kalshi.	
log_analysis_note	market_id, note_text	Confirmation that a note/analysis record was created (Student 3/4 scope, stored locally).	Student 1
Validate_portfolio	portfolio_id (integer 1–100000)	valid flag, errors, warnings (duplicates, self-hedge, > 50% concentration), cost basis, max payout, max loss, max profit	Student 2 · S2
Get_exposure_summary	none	Total exposure and share per market segment	Student 2 · S2
Compare_markets	Polymarketid, kalshi_ticker	Both markets and their price gap, with the basis used (order book or last trade)	Student 4 · S4

5.2 Tool Boundaries

- Registry: Only tools in TOOLS run; anything else returns HTTP 400 with the available list.
- Per-feature allowlists: Student 2 and Student 4 backends reject tools outside their set before any call; Students 1 and 3 call one fixed tool.
- Read-only data access: portfolio tools only send GET requests to the student 2 API. Market tools only read public exchange data.
- Input validation: missing or wrongly typed parameters return a structured error, never an exception.

5.3 RAG Knowledge Sources

The team has decided to use live data from two public, no-authentication APIs, Polymarket and Kalshi, rather than seeded/mock market data, to ground RAG responses in real, current markets. Kalshi is not used as a live source, since it requires registration and is a regulated exchange; this keeps the live-data scope to two openly accessible APIs.

- Live market data (title, category, current probability, volume) fetched from the Polymarket public API, cached locally to protect against rate limits and downtime.
- Live Polymarket and Kalshi markets (pinned IDs, max 3, or best keyword match), The loader is `load_notes_entries`.
- Research notes and tagged evidence (from Student 3's Market Research Notes & News Feed database).
- News snippets linked to markets (from the same feature).
- Retrieval is keyword overlap, not embeddings; section 9 records the trade-off. A loader whose service is down returns nothing, so the server keeps working.

5.4 Confidence Category Definition

Confidence is computed from retrieval, not the model.

Category	Definition
----------	------------

High	2 or more citations
Medium	Exactly 1 citation
Low	0 citations: Fixed message, insufficient_context: true, no LLM call

6. Validation and Results

Release 1 was validated at five levels, from automated tests to the browser; all 68 automated tests pass with MCP and RAG disabled, and the screenshots below show each feature's MCP and RAG interaction on the running stack.

6.1 Validation Approach

Each feature's MCP and RAG integration is validated in two ways: (1) direct terminal/curl testing against the local MCP and RAG servers, isolating the server from the frontend/backend; and (2) end-to-end testing through the actual frontend UI, confirming the full request chain works as experienced by a user. Below is the key components involved in the approach:

1. Automated (CI): pytest per feature plus the shared data layer and compare_markets tests, with MCP and RAG disabled; Docker image build per feature.
2. Terminal: direct calls to GET /health, POST /mcp/call and POST /rag/query, isolating each server from the features.
3. Backend/API: direct calls to each feature's MCP and RAG routes, checking the JSON each frontend receives.
4. Frontend (end to end): each panel used in the browser on the running Compose stack.
5. Agentic loop: python ai-services/agentic-loop/loop.py --mode mcp and --mode rag, with the JSON logs kept as evidence.

6.2 Per-Feature Validation Evidence

Student 1 Feature – Market Watchlist and Discovery:

MCP Frontend:

Live market search (MCP)

election

Search Live

polymarket	12%	Will Trump endorse Likud in the 2026 Israeli Election?
polymarket	69%	Will Democrats sweep the Alaska Senate and Governor elections?
polymarket	84%	Will Trump not endorse a Party in the 2026 Israeli Election?
polymarket	56%	Will Democrats sweep the Maine Senate and Governor elections?
polymarket	89%	Will Democrats sweep the Minnesota Senate and Governor elections?
polymarket	55%	Will Republicans sweep the Kansas Senate and Governor elections?
polymarket	72%	Will Democrats sweep the Michigan Senate and Governor elections?
polymarket	43%	Will Democrats sweep the Iowa Senate and Governor elections?
polymarket	46%	Will Democrats sweep the Georgia Senate and Governor elections?
polymarket	43%	Will Democrats win the Iowa Governor election and Republicans win the Iowa Senate election?

MCP Backend:

```

PS C:\Users\camel\OneDrive\Desktop\ADV Software Dev\updated\ADVSD-main(4)\ADVSD-main> Invoke-RestMethod -Method POST -Uri "http://localhost:8100/mcp/call" -ContentType "application/json" -Body '{"
tool":"search_markets","params":{"query":"Bitcoin","source":"polymarket"}}' | ConvertTo-Json -Depth 10
>>
{
  "meta": {
    "errors": [
      ],
    "fresh": true,
    "source": "live"
  },
  "result": [
    {
      "category": "Bitcoin",
      "close_date": "2027-01-01T05:00:00Z",
      "id": "701502",
      "probability": 0.037,
      "relevance": 1.0,
      "source": "polymarket",
      "title": "Will Bitcoin dip to $45,000 by December 31, 2026?",
      "volume": 8190576.211632003,
      "yes_ask": 0.038,
      "yes_bid": 0.036
    },
    {
      "category": "Bitcoin",
      "close_date": "2027-01-01T05:00:00Z",
      "id": "701501",
      "probability": 0.075,
      "relevance": 1.0,
      "source": "polymarket",
      "title": "Will Bitcoin dip to $55,000 by December 31, 2026?",
      "volume": 6510464.076647001,
      "yes_ask": 0.08,
      "yes_bid": 0.07
    },
    {
      "category": "Bitcoin",
      "close_date": "2027-01-01T05:00:00Z",
      "id": "1057883",
      "probability": 0.0065,
      "relevance": 1.0,
      "source": "polymarket",
      "title": "Will Bitcoin reach $250,000 by December 31, 2026?",
      "volume": 5493926.195621007,
      "yes_ask": 0.007,
      "yes_bid": 0.006
    },
    {
      "category": "Bitcoin",
      "close_date": "2027-01-01T05:00:00Z",
      "id": "1057916",
      "probability": 0.0065,
      "relevance": 1.0,
      "source": "polymarket",
      "title": "Will Bitcoin dip to $15,000 by December 31, 2026?",
      "volume": 5048491.2495230045,
      "yes_ask": 0.007,
      "yes_bid": 0.006
    },
    {
      "category": "Bitcoin",
      "close_date": "2027-01-01T05:00:00Z",
      "id": "1339767",
      "probability": 0.055,
      "relevance": 1.0,
      "source": "polymarket",
      "title": "Will Bitcoin dip to $50,000 by December 31, 2026?",
      "volume": 4140345.4891130007,
      "yes_ask": 0.06,
      "yes_bid": 0.05
    },
    {
      "category": "Bitcoin",
      "close_date": "2027-01-01T05:00:00Z",
      "id": "701496",
      "probability": 0.335,

```

```

    "relevance": 1.0,
    "source": "polymarket",
    "title": "Will Bitcoin reach $100,000 by December 31, 2026?",
    "volume": 3740446.906951001,
    "yes_ask": 0.34,
    "yes_bid": 0.33
  },
  {
    "category": "Bitcoin",
    "close_date": "2027-01-01T05:00:00Z",
    "id": "701503",
    "probability": 0.0215,
    "relevance": 1.0,
    "source": "polymarket",
    "title": "Will Bitcoin dip to $35,000 by December 31, 2026?",
    "volume": 3428304.055725999,
    "yes_ask": 0.022,
    "yes_bid": 0.021
  },
  {
    "category": "Bitcoin",
    "close_date": "2027-01-01T05:00:00Z",
    "id": "1393070",
    "probability": 0.0025,
    "relevance": 1.0,
    "source": "polymarket",
    "title": "Will Bitcoin reach $1,000,000 by December 31, 2026?",
    "volume": 2851358.2673639995,
    "yes_ask": 0.003,
    "yes_bid": 0.002
  },
  {
    "category": "Bitcoin",
    "close_date": "2027-01-01T05:00:00Z",
    "id": "1339768",
    "probability": 0.0305,
    "relevance": 1.0,
    "source": "polymarket",
    "title": "Will Bitcoin dip to $40,000 by December 31, 2026?",
    "volume": 2530241.5373429996,
    "yes_ask": 0.031,
    "yes_bid": 0.03
  },
  {
    "category": "Bitcoin",
    "close_date": "2027-01-01T05:00:00Z",
    "id": "1345531",
    "probability": 0.695,
    "relevance": 1.0,
    "source": "polymarket",
    "title": "Will Bitcoin reach $90,000 by December 31, 2026?",
    "volume": 2259917.6311430004,
    "yes_ask": 0.7,
    "yes_bid": 0.69
  }
}
"tool": "search_markets"
}
PS C:\Users\came1\OneDrive\Desktop\ADV Software Dev\updated\ADVSD-main(4)\ADVSD-main>

```

RAG Frontend:

Ask a grounded question (RAG)

what's driving sentiments in election markets

Ask

The driving force behind tightening election markets and changing Kalshi prices is likely the recent polling aggregates showing a tightening race.

Confidence: High

- [placeholder-knowledge-base] Placeholder note: polling aggregates have shown a tightening race in recent weeks.
- [kalshi] Kalshi: WI-06 House winner? — Brad Smith — probability 0.063 (bid 5.6c, ask 7.0c), volume 21529

RAG Backend:

```

PS C:\Users\camel\OneDrive\Desktop\ADV Software Dev\updated\ADVSD-main(4)\ADVSD-main> Invoke-RestMethod -Method POST -Uri "http://localhost:8200/rag/query" -ContentType "application/json" -Body '{"question":"What is driving the tightening election markets?"}' | ConvertTo-Json -Depth 10
{
  "answer": "The context does not provide a clear answer to what is driving the tightening election markets, only providing polling aggregate data and a specific election result in WI-06 House, but no underlying reason or explanation for the tightening race.",
  "citations": [
    {
      "excerpt": "Placeholder note: polling aggregates have shown a tightening race in recent weeks.",
      "id": "kb-001",
      "source": "placeholder-knowledge-base"
    },
    {
      "excerpt": "Kalshi: WI-06 House winner? – Brad Smith – probability 0.066 (bid 5.6c, ask 7.6c), volume 21529",
      "id": "market:KXHOUSERACE-WI06-26-D",
      "source": "kalshi"
    }
  ],
  "confidence": "High",
  "insufficient_context": false
}
PS C:\Users\camel\OneDrive\Desktop\ADV Software Dev\updated\ADVSD-main(4)\ADVSD-main>

```

Student 2 Feature – Portfolio and Position Tracker:

MCP Frontend:

Portfolio Validation

Main Portfolio

Run Portfolio Validation Tool

Exposure by Category

MCP tool validate_portfolio returned:

Portfolio 1: VALID · 4 positions checked

Cost basis\$660.0

Max payout\$2200

Max loss\$660.0

Max profit\$1540.0

No issues found.

Raw JSON

```
{
  "errors": [],
  "portfolio_id": 1,
  "positions_checked": 4,
  "totals": {
    "cost_basis": 660.0,
    "max_loss": 660.0,
    "max_payout": 2200,
    "max_profit": 1540.0
  },
  "valid": true,
  "warnings": []
}
```

MCP Backend:

```

c093b1e..a8eb83  main -> main
● PS C:\Users\chris\ADVSD> irm -Method Post http://localhost:5002/mcp/execute -ContentType 'application/json' -Body
'{"tool": "validate_portfolio", "parameters": {"portfolio_id": 2}}' | ConvertTo-Json -Depth 5
{
  "result": {
    "errors": [
      ],
    "portfolio_id": 2,
    "positions_checked": 2,
    "totals": {
      "cost_basis": 38.75,
      "max_loss": 38.75,
      "max_payout": 125,
      "max_profit": 86.25
    },
    "valid": true,
    "warnings": [
      "#4 CRYPTO-BTC-100K: 52% of portfolio cost (limit 50%)"
    ]
  },
  "status": "ok",
  "tool": "validate_portfolio"
}
○ PS C:\Users\chris\ADVSD>

```

RAG Frontend:

Market Intelligence

Why is my POL-TRUMP-2026 position down?

Query Knowledge Base

Answer: The POL-TRUMP-2026 position is down because it was initially set up to go long, but you sold 20 contracts at \$0.58, which is below the buy price of \$0.55, resulting in an unrealised P&L of -\$1.0.

Confidence: High

Sources:

- [S1]** student-2-portfolio · portfolio:position#1
Portfolio position POL-TRUMP-2026 YES (Politics 2026): 100 contracts bought at \$0.55 (cost \$55.00, max payout \$100). Price moved up from 0.55 on 2026-08-31 to 0.56 on 2026-09-29 (low 0.49, high 0.62); unrealised P&L \$1.0. Trades: SELL 20 @ \$0.58, BUY 100 @ \$0.55.
- [S2]** student-2-portfolio · portfolio:exposure-summary
Portfolio exposure: 10 open positions, total cost \$853.75. By category: KALSHI \$380.00 (45%), ECON \$180.00 (21%), POL \$155.00 (18%), SPORTS \$100.00 (12%), CRYPTO \$38.75 (5%). Holdings: POL-TRUMP-2026 YES x100 @ \$0.55; KALSHI-FED-SEP NO x500 @ \$0.2; POL-BIDEN-2026 NO x200 @ \$0.15; CRYPTO-BTC-100K YES x50 @ \$0.4; POL-HARRIS-2026 YES x150 @ \$0.45; ECON-INFLATION-UP YES x300 @ \$0.6; SPORTS-NFL-SB NO x1000 @ \$0.1; POL-NEWSOM-2026 NO x50 @ \$0.05; CRYPTO-ETH-10K YES x75 @ \$0.25; KALSHI-RATES-DOWN YES x400 @ \$0.7.

RAG Backend:

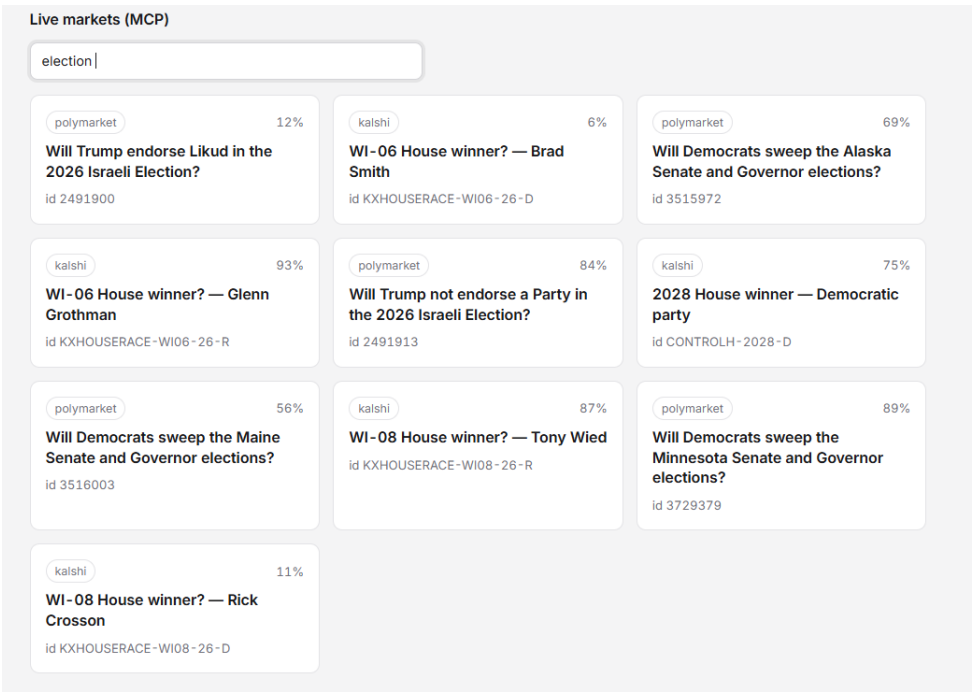
```

}
● PS C:\Users\chris\ADVSD> irm -Method Post http://localhost:5002/rag/ask -ContentType 'application/json' -Body '{"query": "Why is my POL-TRUMP-2026 position down?"}' | ConvertTo-Json -Depth 5
{
  "answer": "The POL-TRUMP-2026 position is down because it was sold at $0.58, which is lower than the original buy price of $0.55, resulting in an unrealized loss of $0.03 per contract.",
  "citations": [
    {
      "id": "S1",
      "ref": "portfolio:position#1",
      "snippet": "Portfolio position POL-TRUMP-2026 YES (Politics 2026): 100 contracts bought at $0.55 (cost $55.00, max payout $100). Price moved up from 0.55 on 2026-08-31 to 0.56 on 2026-09-29 (low 0.49, high 0.62); unrealised P\0026L $1.0. Trades: SELL 20 @ $0.58, BUY 100 @ $0.55.",
      "source": "student-2-portfolio"
    },
    {
      "id": "S2",
      "ref": "portfolio:exposure-summary",
      "snippet": "Portfolio exposure: 10 open positions, total cost $853.75. By category: KALSHI $380.00 (45%), ECON $180.00 (21%), POL $155.00 (18%), SPORTS $100.00 (12%), CRYPTO $38.75 (5%). Holdings: POL-TRUMP-2026 YES x100 @ $0.55; KALSHI-FED-SEP NO x500 @ $0.2; POL-BIDEN-2026 NO x200 @ $0.15; CRYPTO-BTC-100K YES x50 @ $0.4; POL-HARRIS-2026 YES x150 @ $0.45; ECON-INFLATION-UP YES x300 @ $0.6; SPORTS-NFL-SB NO x1000 @ $0.1; POL-NEWSOM-2026 NO x50 @ $0.05; CRYPTO-ETH-10K YES x75 @ $0.25; KALSHI-RATES-DOWN YES x400 @ $0.7.",
      "source": "student-2-portfolio"
    }
  ],
  "confidence": "High",
  "generated": true,
  "status": "ok"
}
○ PS C:\Users\chris\ADVSD>

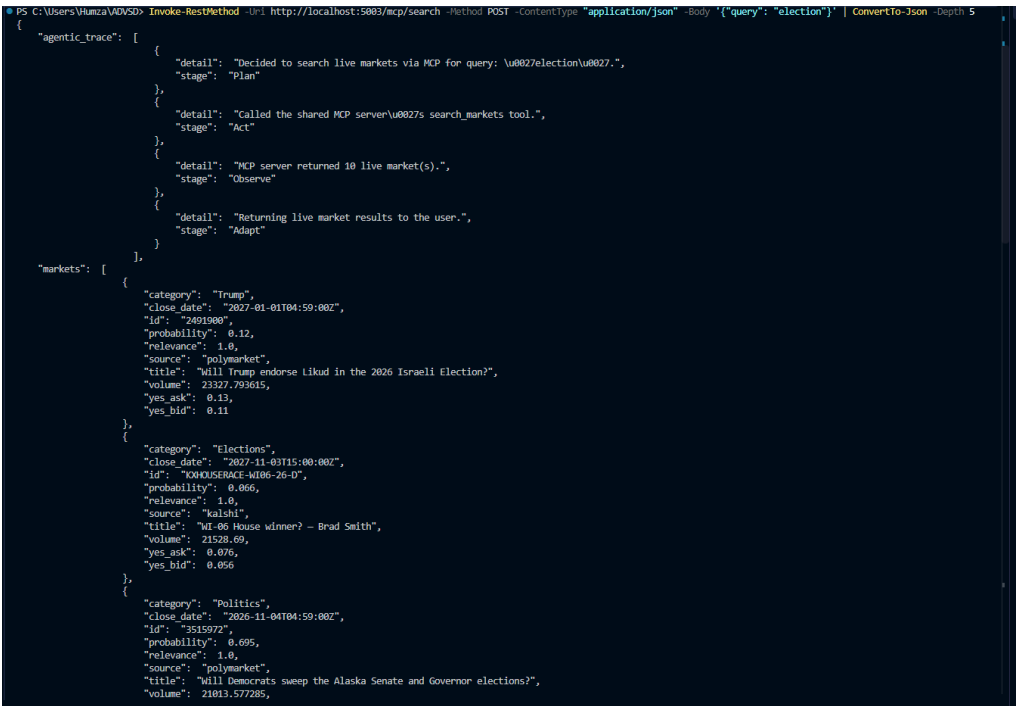
```

Student 3 feature – Market Research Notes and News Feed:

MCP Frontend:



MCP Backend:



RAG Frontend:

Ask, grounded in your notes (RAG)

whats the latest on the election

Ask

Election polling has tightened in key swing states, with recent surveys showing a narrowing race, according to a news report from August 18, 2026.

Confidence: High

- placeholder-knowledge-base: Placeholder note: polling aggregates have shown a tightening race in recent weeks.
- student-2-portfolio: Most recently opened position: KALSHI-RATES-DOWN YES, 400 contracts at \$0.7 (cost \$280.00, max payout \$400).
- student-3-news-feed: News: Election polling tightens in key swing states (Politics, 2026-08-18) — New surveys show a narrowing race with week

RAG Backend:

```

}
Invoke-WebRequest -Uri http://localhost:5003/rag/ask -Method POST -ContentType "application/json" -Body '{"question": "pick a question matching one of your seed
ed notes or news articles"}' | ConvertTo-Json -Depth 5
{
  "agentic_trace": [
    {
      "detail": "Decided to ask the shared RAG server, grounded in research notes and news: \u0027pick a question matching one of your seeded notes or news
articles\u003e\u0027.",
      "stage": "Plan"
    },
    {
      "detail": "Called the shared RAG server\u0027s /rag/query endpoint.",
      "stage": "Act"
    },
    {
      "detail": "RAG server returned an answer with confidence=Low, insufficient_context=True.",
      "stage": "Observe"
    },
    {
      "detail": "Returned the grounded answer, citations, and confidence to the user.",
      "stage": "Adapt"
    }
  ],
  "answer": "I don\u0027t have enough relevant context to answer that confidently. Try asking about a specific market or a topic covered in our research notes.",
  "citations": [
  ],
  "confidence": "Low",
  "insufficient_context": true
}

```

Student 4 Feature – AI Market Analyst Assistant:

MCP Frontend:

Live data via the shared MCP server. Pick the same event on each exchange, then compare. Mispriced means the order books don't overlap: one exchange's bid is above the other's ask.

Polymarket

- ☐ US recession by end of 2026? 8.5% · bid 8.0¢ / ask 9.0¢
- ☐ UK Recession in 2026? 9.5% · bid 9.0¢ / ask 10.0¢
- ☒ US recession by end of 2027? 34.5% · bid 34.0¢ / ask 35.0¢
- ☐ Japan recession in 2026? 4.5% · bid 4.0¢ / ask 5.0¢

Kalshi

- ☐ Recession this year? — Starts 4.5% · bid 4.0¢ / ask 5.0¢
- ☒ Recession in 2027? — Yes 21.5% · bid 21.0¢ / ask 22.0¢
- ☐ When will the next US recession start? — Q2 2025 1.8% · bid 0.1¢ / ask 3.4¢
- ☐ When will the next US recession start? — Q4 2025 3.6% · bid 0.6¢ / ask 6.6¢
- ☐ When will the next US recession start? — Q1 2026 5.9% · bid 3.1¢ / ask 8.8¢

Compare selected pair

Plan → Act → Observe → Adapt

MISPRICED Polymarket prices YES higher than Kalshi

Polymarket - YES	Kalshi - YES	Gap
34.5% bid 34.0¢ · ask 35.0¢ US recession by end of 2027? 3407418	21.5% bid 21.0¢ · ask 22.0¢ Recession in 2027? — Yes KXRECSNBER-27	13.0 pts edge after spread 12.0 pts Compared on order books (bid/ask).

Save as analysis **Explain this gap (RAG)**

Plan → Act → Observe → Adapt

```
PS C:\Users\Erf\Documents\Github\VASDV\VASDV> Invoke-RestMethod -Method POST -Uri "http://localhost:5004/mcp/execute" -ContentType "application/json" -Body '{"tool":"compare_markets","parameters":{"polymarket_id":"3407418","kalshi_ticker":"KXRECSNBER-27"}}' | ConvertTo-Json -Depth 10
{
  "agentic_trace": [
    {
      "detail": "Selected MCP tool \u002fcompare_markets\u002f with parameters {\u002fpolymarket_id\u002f: \u002f3407418\u002f, \u002fkalshi_ticker\u002f: \u002fKXRECSNBER-27\u002f}.",
      "stage": "Plan"
    },
    {
      "detail": "Called the shared MCP server\u002f\u002fs /mcp/call endpoint.",
      "stage": "Act"
    },
    {
      "detail": "Shared MCP server returned a structured result.",
      "stage": "Observe"
    },
    {
      "detail": "Returned the tool result to the frontend.",
      "stage": "Adapt"
    }
  ],
  "meta": {
  },
  "result": {
    "comparison": {
      "abs_gap_points": 17.0,
      "basis": "order books (bid/ask)",
      "direction": "Polymarket prices YES higher than Kalshi",
      "edge_after_spread": 0.16,
      "gap": 0.17,
      "status": "mispriced"
    },
    "kalshi": {
      "category": "Economics",
      "close_date": "2028-01-31T13:25:00Z",
      "id": "KXRECSNBER-27",
      "probability": 0.165,
      "source": "Kalshi",
      "title": "Recession in 2027? - Yes",
      "volume": 630388.49,
      "yes_ask": 0.17,
      "yes_bid": 0.16
    },
    "polymarket": {
      "category": "General",
```

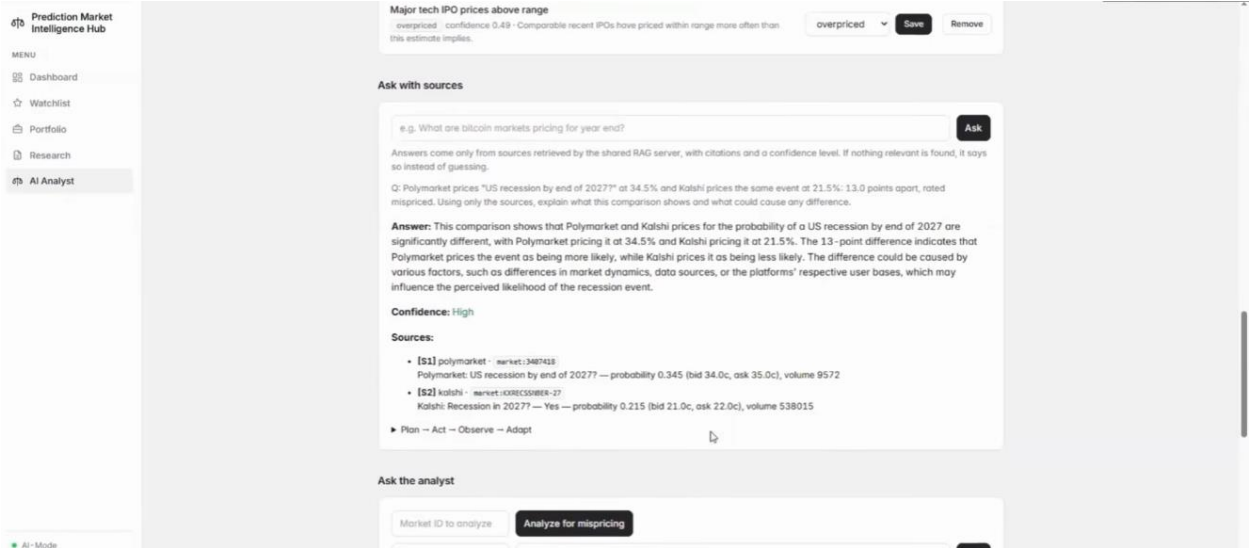
Activate Windows
Go to Settings to activate Windows.

```
    "meta": {
      },
    },
    "result": {
      "comparison": {
        "abs_gap_points": 17.0,
        "basis": "order books (bid/ask)",
        "direction": "Polymarket prices YES higher than Kalshi",
        "edge_after_spread": 0.16,
        "gap": 0.17,
        "status": "mispriced"
      },
      "kalshi": {
        "category": "Economics",
        "close_date": "2028-01-31T13:25:00Z",
        "id": "KXRECSNBER-27",
        "probability": 0.165,
        "source": "Kalshi",
        "title": "Recession in 2027? - Yes",
        "volume": 63038.49,
        "yes_ask": 0.17,
        "yes_bid": 0.16
      },
      "polymarket": {
        "category": "General",
        "close_date": "2027-12-31T23:59:00Z",
        "id": "3407418",
        "probability": 0.335,
        "source": "polymarket",
        "title": "US recession by end of 2027?",
        "volume": 10101.66167,
        "yes_ask": 0.34,
        "yes_bid": 0.33
      }
    },
    "status": "ok",
    "tool": "compare_markets"
  }
}
```

S C:\Users\Erf\Documents\Github\ASDV\ADVS\ASDVSD> |

Activate Windows
Go to Settings to activate Windows.

RAG Frontend:



RAG Backend:

```
PS C:\Users\Erf\Documents\Github\ASDV\ADVS> Invoke-RestMethod -Method POST -Uri "http://localhost:5004/rag/ask" -ContentType "application/json" -Body '{"query": "Will the US enter a recession by 2027?", "market_ids": ["3407418", "KXRECSNBER-27"]}' | ConvertTo-Json -Depth 10
{
  "agentic_trace": [
    {
      "detail": "Ask the shared RAG server to retrieve sources and answer only from them, pinned to markets 3407418, KXRECSNBER-27.",
      "stage": "Plan"
    },
    {
      "detail": "Called the shared RAG server\u002Fs /rag/query endpoint.",
      "stage": "Act"
    },
    {
      "detail": "Retrieved 2 source(s); confidence High.",
      "stage": "Observe"
    },
    {
      "detail": "Returned the grounded answer with its citations.",
      "stage": "Adapt"
    }
  ],
  "answer": "There is no definitive answer to this question based on the provided context.",
  "citations": [
    {
      "id": "51",
      "ref": "market:3407418",
      "snippet": "Polymarket: US recession by end of 2027? - probability 0.335 (bid 33.0c, ask 34.0c), volume 10102",
      "source": "polymarket"
    },
    {
      "id": "52",
      "ref": "market:KXRECSNBER-27",
      "snippet": "Kalshi: Recession in 2027? - Yes - probability 0.165 (bid 16.0c, ask 17.0c), volume 630388",
      "source": "kalshi"
    }
  ],
  "confidence": "High",
  "generated": true,
  "status": "ok"
}
PS C:\Users\Erf\Documents\Github\ASDV\ADVS>
PS C:\Users\Erf\Documents\Github\ASDV\ADVS> []
```

6.3 Local Terminal Validation

MCP Server Validation

```
PS C:\Users\chris\ADVS> irm -Method Post http://localhost:8100/mcp/call -ContentType 'application/json' -Body '{"tool": "get_exposure_summary", "params": {}}' | ConvertTo-Json -Depth 5
{
  "result": {
    "by_category": [
      {
        "category": "KALSHI",
        "exposure": 380.0,
        "share": 0.476
      },
      {
        "category": "ECON",
        "exposure": 180.0,
        "share": 0.225
      },
      {
        "category": "POL",
        "exposure": 100.0,
        "share": 0.125
      },
      {
        "category": "SPORTS",
        "exposure": 100.0,
        "share": 0.125
      },
      {
        "category": "CRYPTO",
        "exposure": 38.75,
        "share": 0.049
      }
    ],
    "total_exposure": 798.75
  },
  "tool": "get_exposure_summary"
}
PS C:\Users\chris\ADVS>
```

Rag Terminal Validation

```
PS C:\Users\chris\Documents> irm -Method Post http://localhost:8200/rag/query -ContentType 'application/json' -Body '{"question":"How concentrated is my portfolio exposure?"}' | ConvertTo-Json -Depth 5
{
  "answer": "Your portfolio exposure appears to be moderately concentrated, with a significant portion ($380.00, 48%) allocated to a single category (KALSHI).",
  "citations": [
    {
      "excerpt": "Portfolio exposure: 9 open positions, total cost $798.75. By category: KALSHI $380.00 (48%), ECON $180.00 (23%), POL $100.00 (13%), SPORTS $100.00 (13%), CRYPTO $38.75 (5%). Holdings: KALSHI-FED-SEP NO x500 @ $0.2; POL-BIDEN-2026 NO x200 @ $0.15; CRYPTO-BTC-100K YES x50 @ $0.4; POL-HARRIS-2026 YES x150 @ $0.45; ECON-INFLATION-UP YES x300 @ $0.6; SPORTS-NFL-SB N O x1000 @ $0.1; POL-NEWSOM-2026 NO x50 @ $0.05; CRYPTO-ETH-10K YES x75 @ $0.25; KALSHI-RATES-DOWN YES x800 @ $0.7.",
      "id": "portfolio:exposure-summary",
      "source": "student-2-portfolio"
    },
    {
      "excerpt": "Research note on \u0022Central bank signals possible rate move\u0022: Rate move implications - Watch for volatility in rate-sensitive markets. (tags: macro,rates)",
      "id": "student3-note#1",
      "source": "student-3-research-notes"
    },
    {
      "excerpt": "News: Central bank signals possible rate move (Economics, 2026-08-20) - Officials hint at policy shift at next meeting, markets react.",
      "id": "student3-news#1",
      "source": "student-3-news-feed"
    }
  ],
  "confidence": "High",
  "insufficient_context": false
}
```

6.4 Grounded Response and Insufficient-Context Examples

Question	Result
“Why is my POL-TRUMP-2026 position down?”	Grounded answer, Medium confidence, citation portfolio:position#1 with its price history and trades
“How concentrated is my portfolio exposure?”	Grounded answer citing portfolio:exposure-summary
“What is the weather in Tokyo?”	insufficient_context: true, no citations, fixed refusal message, no LLM call

6.5 Shared Agentic Loop Outputs

MCP Server Validation

```

PS C:\Users\chris\ADVSD> python ai-services/agentlc-loop/loop.py --mode mcp
=== Agentic Loop: MCP Validation Mode ===
[Plan] Decided to validate the MCP server by calling 'search_markets' with {'query': 'election', 'source': 'both'}.
[Act] Calling MCP server at http://localhost:8100/mcp/call
[Observe] MCP server returned a structured result.
[Adapt] MCP validation passed: tool call returned a structured result.

--- Raw result ---
{
  "meta": {
    "errors": [],
    "fresh": true,
    "source": "live"
  },
  "result": [
    {
      "category": "Trump",
      "close_date": "2027-01-01T04:59:00Z",
      "id": "2491900",
      "probability": 0.12,
      "relevance": 1.0,
      "source": "polymarket",
      "title": "Will Trump endorse Likud in the 2026 Israeli Election?",
      "volume": 23327.793615000002,
      "yes_ask": 0.13,
      "yes_bid": 0.11
    },
    {
      "category": "Elections",
      "close_date": "2027-11-03T15:00:00Z",
      "id": "KXHOUSEACE-WI06-26-D",
      "probability": 0.066,
      "relevance": 1.0,
      "source": "kalshi",
      "title": "WI-06 House winner? \u2014 Brad Smith",
      "volume": 21528.69,
      "yes_ask": 0.076,
      "yes_bid": 0.056
    }
  ]
}

```

```

{
  "category": "Politics",
  "close_date": "2026-11-04T04:59:00Z",
  "id": "3515972",
  "probability": 0.695,
  "relevance": 1.0,
  "source": "polymarket",
  "title": "Will Democrats sweep the Alaska Senate and Governor elections?",
  "volume": 21013.577285,
  "yes_ask": 0.7,
  "yes_bid": 0.69
},
{
  "category": "Elections",
  "close_date": "2027-11-03T15:00:00Z",
  "id": "KXHOUSERACE-WI06-26-R",
  "probability": 0.9315,
  "relevance": 1.0,
  "source": "kalshi",
  "title": "WI-06 House winner? \u2014 Glenn Grothman",
  "volume": 19518.6,
  "yes_ask": 0.941,
  "yes_bid": 0.922
},
{
  "category": "Trump",
  "close_date": "2027-01-01T04:59:00Z",
  "id": "2491913",
  "probability": 0.84,
  "relevance": 1.0,
  "source": "polymarket",
  "title": "Will Trump not endorse a Party in the 2026 Israeli Election?",
  "volume": 18103.152923,
  "yes_ask": 0.85,
  "yes_bid": 0.83
},
{

```

```

{
  "category": "Elections",
  "close_date": "2029-02-01T15:00:00Z",
  "id": "CONTROLH-2028-D",
  "probability": 0.75,
  "relevance": 1.0,
  "source": "kalshi",
  "title": "2028 House winner \u2014 Democratic party",
  "volume": 15408.89,
  "yes_ask": 0.76,
  "yes_bid": 0.74
},
{
  "category": "Politics",
  "close_date": "2026-11-04T04:59:00Z",
  "id": "3516003",
  "probability": 0.565,
  "relevance": 1.0,
  "source": "polymarket",
  "title": "Will Democrats sweep the Maine Senate and Governor elections?",
  "volume": 14684.405548,
  "yes_ask": 0.57,
  "yes_bid": 0.56
},
{
  "category": "Elections",
  "close_date": "2027-11-03T15:00:00Z",
  "id": "KXHOUSERACE-WI08-26-R",
  "probability": 0.875,
  "relevance": 1.0,
  "source": "kalshi",
  "title": "WI-08 House winner? \u2014 Tony Wied",
  "volume": 12830.65,
  "yes_ask": 0.88,
  "yes_bid": 0.87
},
{

```

```

},
{
  "category": "Politics",
  "close_date": "2026-11-04T04:59:00Z",
  "id": "3729379",
  "probability": 0.895,
  "relevance": 1.0,
  "source": "polymarket",
  "title": "Will Democrats sweep the Minnesota Senate and Governor elections?",
  "volume": 12842.898793999999,
  "yes_ask": 0.9,
  "yes_bid": 0.89
},
{
  "category": "Elections",
  "close_date": "2027-11-03T15:00:00Z",
  "id": "KXHOUSERACE-WI08-26-D",
  "probability": 0.115,
  "relevance": 1.0,
  "source": "kalshi",
  "title": "WI-08 House winner? \u2014 Rick Crosson",
  "volume": 8019.94,
  "yes_ask": 0.13,
  "yes_bid": 0.1
}
],
"tool": "search_markets"

```

Rag Terminal Validation

```
PS C:\Users\chris\ADVSD> python ai-services\agentic-loop\loop.py --mode rag
=== Agentic loop: RAG Validation Mode ===
[Plan] Decided to validate the RAG server with a grounded question: 'What is driving sentiment in election markets right now?'
[Act] calling RAG server at http://localhost:8200/rag/query
[Observe] RAG server returned an answer with confidence-High, 4 citation(s), insufficient_context-False.
[Adapt] RAG validation passed: grounded answer with citations and a confidence category was returned.

--- Raw result ---
{
  "answer": "Election markets indicate tightening races in key swing states, which is driving sentiment and narrowing the gap in the polls.",
  "citations": [
    {
      "excerpt": "Placeholder note: polling aggregates have shown a tightening race in recent weeks.",
      "id": "kb-001",
      "source": "placeholder-knowledge-base"
    },
    {
      "excerpt": "Research note on 'Election polling tightens in key swing states': Swing state tracker \u2014 Polling error history suggests wide confidence interval. (tags: po
      litics,polling)",
      "id": "student3-note#3",
      "source": "student-3-research-notes"
    },
    {
      "excerpt": "News: Election polling tightens in key swing states (Politics, 2026-08-18) \u2014 New surveys show a narrowing race with weeks to go.",
      "id": "student3-news#3",
      "source": "student-3-news-feed"
    },
    {
      "excerpt": "Polymarket: Will Trump endorse Likud in the 2026 Israeli Election? \u2014 probability 0.120 (bid 11.0c, ask 13.0c), volume 23328",
      "id": "market:2491900",
      "source": "polymarket"
    }
  ],
  "confidence": "High",
  "insufficient_context": false
}

[saved evidence log: C:\Users\chris\ADVSD\ai-services\agentic-loop\validation_logs\rag_validation_20261007_193224.json]
```

6.6 GitHub Actions Workflow Evidence

Screenshots are in chronological order for each workflow:

Student-1-ci:

student-1-ci
student-1.yml

22 workflow runs

This workflow has a workflow_dispatch event trigger. Run workflow

Merge pull request #16 from Erfanur1/feature/ui-refresh student-1-ci #22: Commit 45970b9 pushed by Erfanur1	main	Oct 1, 12:32 AM GMT+10 20s	...
UI refresh: minimal light theme, shared sidebar, dashboard home student-1-ci #21: Pull request #16 opened by Erfanur1	feature/ui-refresh	Oct 1, 12:29 AM GMT+10 28s	...
UI refresh: version theme.css/nav.js links so browsers don't mix cach... student-1-ci #20: Commit 4b1822b pushed by Erfanur1	feature/ui-refresh	Oct 1, 12:29 AM GMT+10 24s	...
Merge pull request #13 from Erfanur1/release-1/student-1 student-1-ci #19: Commit 0b73445 pushed by AlexChoi2004	main	Sep 29, 12:26 PM GMT+10 25s	...
RRelease 1: Add MCP and RAG integration for Student 1 student-1-ci #18: Pull request #13 opened by AlexChoi2004	release-1/student-1	Sep 29, 12:26 PM GMT+10 22s	...
Release 1: integrate MCP and RAG into Student 1 feature, update CI wo... student-1-ci #17: Commit 2f1eb53 pushed by AlexChoi2004	release-1/student-1	Sep 29, 12:20 PM GMT+10 25s	...

Brief Descriptions:

1. Add MCP and RAG: Added MCP and RAG to the Student 1 feature.
2. Update Student 1 CI: Updated student-1.yml for the Release 1 changes.

3. Commit Release 1 Changes: Committed the MCP/RAG changes to the `release-1/student-1` branch (2f1eb53).
4. Open Pull Request: Opened pull request #13 for the Student 1 Release 1 changes.
5. Merge Pull Request #13: Merged the Release 1 changes into `main` (0b73445).
6. Check CI Still Works: Confirmed the Student 1 CI workflow still runs successfully after the Release 1 changes.

Student-2-ci:

student-2-ci

student-2.yml

Filter workflow runs

...

12 workflow runs

Event ▾ Status ▾ Branch ▾ Actor ▾

This workflow has a workflow_dispatch event trigger.

Run workflow ▾

✓

student-2: render portfolio dropdown server-side

student-2-ci #12: Commit ba4a7a1 pushed by christopherkumarellicherian-bit

main

🕒

Oct 1, 10:48 PM GMT+10

🕒

22s

...

✓

Merge pull request #16 from Erfanur1/feature/ui-refresh

student-2-ci #11: Commit 45970b9 pushed by Erfanur1

main

🕒

Oct 1, 12:32 AM GMT+10

🕒

29s

...

✓

UI refresh: minimal light theme, shared sidebar, dashboard home

student-2-ci #10: Pull request #16 opened by Erfanur1

feature/ui-refresh

🕒

Oct 1, 12:29 AM GMT+10

🕒

27s

...

✓

UI refresh: version theme.css/nav.js links so browsers don't mix cach...

student-2-ci #9: Commit 4b1822b pushed by Erfanur1

feature/ui-refresh

🕒

Oct 1, 12:29 AM GMT+10

🕒

36s

...

✓

compose: point student-2 backend at shared MCP (8100) and RAG (8200)

student-2-ci #8: Commit a8ebe83 pushed by christopherkumarellicherian-bit

main

🕒

Sep 30, 2:33 AM GMT+10

🕒

25s

...

✓

Merge main into local

student-2-ci #7: Commit c093b1e pushed by christopherkumarellicherian-bit

main

🕒

Sep 29, 2:04 PM GMT+10

🕒

22s

...

Brief Descriptions:

1. render portfolio dropdown server-side (ba4a7a1, 1 Oct, 22s): moved the portfolio dropdown so the server fills it in when the page loads, which makes it reliable. Tests and the Docker build passed with MCP/RAG disabled.
2. Merge pull request #16 (feature/ui-refresh) (4397db9, 1 Oct, 29s): Erfanur's shared UI refresh merged into main. Re-ran the Student 2 checks to confirm the new theme didn't break the feature.
3. UI refresh: minimal light theme, shared sidebar, dashboard home (PR #16, 1 Oct, 27s): checked the shared theme changes against the Student 2 build before merging.
4. version theme.css/nav.js links (4b1822b, 1 Oct, 26s): added version numbers to the shared CSS/JS links so browsers don't mix old and new cached files. Student 2 build and tests still pass.
5. point student-2 backend at shared MCP (8100) and RAG (8200) (a8ebe83, 30 Sep, 25s): updated the Docker Compose connection settings so the Student 2 backend reaches the shared MCP and RAG servers on the host. CI passed with `MCP_ENABLED=false` and `RAG_ENABLED=false`.
6. Merge main into local (c093b1e, 29 Sep, 22s): merged teammates' changes into the price-history chart work. The full test suite and Docker image build passed after the merge.

Student-3-ci:

student-3-ci

student-3.yml

Filter workflow runs

...

19 workflow runs				Event	Status	Branch	Actor
This workflow has a workflow_dispatch event trigger.				Run workflow			
✓	Merge pull request #18 from Erfanur1/fix/student-3-insufficient-context	main	Oct 1, 9:15 PM GMT+10	...			
	student-3-ci #19: Commit a70221a pushed by SHumzaHussain		25s				
✓	fixed insufficient context	fix/student-3-insufficient-	Oct 1, 9:13 PM GMT+10	...			
	student-3-ci #18: Pull request #18 opened by SHumzaHussain		36s				
✓	fixed insufficient context	fix/student-3-insufficient-	Oct 1, 9:13 PM GMT+10	...			
	student-3-ci #17: Commit 7e087cf pushed by SHumzaHussain		28s				
✓	Merge pull request #17 from Erfanur1/feature/student-3-release-1	main	Oct 1, 8:38 PM GMT+10	...			
	student-3-ci #16: Commit 5ae6b99 pushed by SHumzaHussain		25s				
✓	Merging release 1 update for Student-3 Feature-3. Resovled conflict w...	feature/student-3-release-1	Oct 1, 8:35 PM GMT+10	...			
	student-3-ci #15: Pull request #17 opened by SHumzaHussain		26s				
✓	Merging release 1 update for Student-3 Feature-3. Resovled conflict w...	feature/student-3-release-1	Oct 1, 8:35 PM GMT+10	...			
	student-3-ci #14: Commit df5192a pushed by SHumzaHussain		28s				
✓	Merge pull request #16 from Erfanur1/feature/ui-refresh	main	Oct 1, 12:32 AM GMT+10	...			
	student-3-ci #13: Commit 45970b9 pushed by Erfanur1		29s				
✓	UI refresh: minimal light theme, shared sidebar, dashboard home	feature/ui-refresh	Oct 1, 12:29 AM GMT+10	...			
	student-3-ci #12: Pull request #16 opened by Erfanur1		20s				
✓	UI refresh: version theme.css/nav.js links so browsers don't mix cach...	feature/ui-refresh	Oct 1, 12:29 AM GMT+10	...			
	student-3-ci #11: Commit 4b1822b pushed by Erfanur1		34s				

Brief Descriptions:

1. Add MCP search and RAG Q&A to the student 3 feature
2. Fixed insufficient context display
3. Open pull request 17 for student 3 release 1 MCP and RAG changes
4. Merge pull request 17
5. Open pull request 18 for insufficient context fix
6. Merge pull request 18 into main

Student-4-ci:

student-4-ci student-4.yml		Filter workflow runs		...
16 workflow runs		Event ▾	Status ▾	Branch ▾ Actor ▾
This workflow has a workflow_dispatch event trigger.		Run workflow ▾		
✓ Merge pull request #16 from Erfanur1/feature/ui-refresh	main	Oct 1, 12:32 AM GMT+10 26s		
student-4-ci #16: Commit 45970b9 pushed by Erfanur1				
✓ UI refresh: minimal light theme, shared sidebar, dashboard home	feature/ui-refresh	Oct 1, 12:29 AM GMT+10 22s		
student-4-ci #15: Pull request #16 opened by Erfanur1				
✓ UI refresh: version theme.css/nav.js links so browsers don't mix cach...	feature/ui-refresh	Oct 1, 12:29 AM GMT+10 31s		
student-4-ci #14: Commit 4b1822b pushed by Erfanur1				
✓ Merge pull request #15 from Erfanur1/feature/student-4-release-1	main	Oct 1, 12:03 AM GMT+10 37s		
student-4-ci #13: Commit cd852a6 pushed by Erfanur1				
✓ Student 4 Release 1: cross-exchange mispricing (MCP) + grounded answers (R...	feature/student-4-release-1	Sep 30, 11:50 PM GMT+10 26s		
student-4-ci #12: Pull request #15 synchronize by Erfanur1				
✓ student-4: state the measured gap in the "Explain this gap" RAG question	feature/student-4-release-1	Sep 30, 11:50 PM GMT+10 29s		
student-4-ci #11: Commit 7fe323f pushed by Erfanur1				
✓ Student 4 Release 1: cross-exchange mispricing (MCP) + grounded answers (R...	feature/student-4-release-1	Sep 30, 11:35 PM GMT+10 25s		
student-4-ci #10: Pull request #15 opened by Erfanur1				
✓ student-4: CI with MCP/RAG disabled; compose MCP/RAG connection	feature/student-4-release-1	Sep 30, 11:34 PM GMT+10 29s		
student-4-ci #9: Commit 28f1cb pushed by Erfanur1				
✓ student-4: CI with MCP/RAG disabled; compose MCP/RAG connection	feature/student-4-release-1	Sep 30, 11:33 PM GMT+10 30s		
student-4-ci #8: Commit acfdeb9 pushed by Erfanur1				

Brief Descriptions:

1. CI with MCP/RAG disabled (acfdeb9): first push of feature/student-4-release-1. student-4.yml sets MCP_ENABLED/RAG_ENABLED=false, runs the feature, data-layer and compare_markets tests, builds the image and smoke-tests the container.
2. CI with MCP/RAG disabled (28f71cb): re-run after rebasing the branch onto the Kalshi data-layer update (PR #14) so it builds on the merged shared code.
3. Open pull request #15: opened the Student 4 Release 1 PR (cross-exchange mispricing via MCP plus grounded answers via RAG).
4. State the measured gap in the RAG question (7fe323f): fixed the LLM making up a price gap when both markets agreed, by putting both prices, the gap and the status into the question.
5. PR #15 synchronised: CI re-ran on the PR with the fix included.
6. Merge pull request #15 (cd852a6): Student 4 Release 1 merged into main. CI passed on main.
7. UI refresh: version theme.css/nav.js links (4b1822b): cache-busting so browsers don't mix the old and new styles.
8. UI refresh: open pull request #16: shared light theme, sidebar and dashboard home across all features.
9. Merge pull request #16 (45970b9): UI refresh merged into main. Student 4 CI still passes.

Each updated student-x.yml workflow retains MCP and RAG integration in the codebase, but both are disabled during CI/CD execution (e.g. via an environment flag or by skipping the relevant test steps), since the MCP and RAG servers are not containerised and are not available in the CI runner.

6.7 Docker Compose Deployment Evidence

```
PS C:\Users\chris\ADVSD> docker compose ps
NAME                IMAGE                COMMAND                SERVICE    CREATED    STATUS    PORTS
advsd-ai-mode-1     advsd-ai-mode       "python app.py"        ai-mode    5 seconds ago    Up 2 seconds    0.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp
advsd-home-1        advsd-home          "python -m http.serv..." home       5 seconds ago    Up 2 seconds    0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
advsd-ollama-1      ollama/ollama:latest "/bin/ollama serve"    ollama     8 days ago      Up 2 seconds    0.0.0.0:11434->11434/tcp, [::]:11434->11434/tcp
advsd-student-1-backend-1  advsd-student-1-backend  "sh -c 'python backe..." student-1-backend  4 seconds ago    Up 2 seconds    0.0.0.0:5001->5001/tcp, [::]:5001->5001/tcp
advsd-student-1-frontend-1  advsd-student-1-frontend  "python frontend/app..." student-1-frontend  4 seconds ago    Up 2 seconds    0.0.0.0:5101->5101/tcp, [::]:5101->5101/tcp
advsd-student-2-backend-1  advsd-student-2-backend  "sh -c 'python backe..." student-2-backend  4 seconds ago    Up 2 seconds    0.0.0.0:5002->5002/tcp, [::]:5002->5002/tcp
advsd-student-2-frontend-1  advsd-student-2-frontend  "python frontend/app..." student-2-frontend  4 seconds ago    Up 2 seconds    0.0.0.0:5102->5102/tcp, [::]:5102->5102/tcp
advsd-student-3-backend-1  advsd-student-3-backend  "sh -c 'python backe..." student-3-backend  4 seconds ago    Up 2 seconds    0.0.0.0:5003->5003/tcp, [::]:5003->5003/tcp
advsd-student-3-frontend-1  advsd-student-3-frontend  "python frontend/app..." student-3-frontend  4 seconds ago    Up 2 seconds    0.0.0.0:5103->5103/tcp, [::]:5103->5103/tcp
advsd-student-4-backend-1  advsd-student-4-backend  "sh -c 'python backe..." student-4-backend  4 seconds ago    Up 2 seconds    0.0.0.0:5004->5004/tcp, [::]:5004->5004/tcp
advsd-student-4-frontend-1  advsd-student-4-frontend  "python frontend/app..." student-4-frontend  4 seconds ago    Up 2 seconds    0.0.0.0:5104->5104/tcp, [::]:5104->5104/tcp
PS C:\Users\chris\ADVSD> docker compose exec student-1-backend printenv MCP_URL RAG_URL
http://host.docker.internal:8100
http://host.docker.internal:8100
PS C:\Users\chris\ADVSD> docker compose exec student-2-backend printenv MCP_SERVER_URL RAG_SERVER_URL
http://host.docker.internal:8100
http://host.docker.internal:8100
PS C:\Users\chris\ADVSD> docker compose exec student-3-backend printenv MCP_URL RAG_URL
http://host.docker.internal:8100
http://host.docker.internal:8200
PS C:\Users\chris\ADVSD> docker compose exec student-4-backend printenv MCP_URL RAG_URL
http://host.docker.internal:8100
http://host.docker.internal:8200
PS C:\Users\chris\ADVSD>
```

6.8 Integration Validation Summary

The Release 1 integration was successfully validated across all four isolated student environments. The introduction of the `host.docker.internal` network gateway allows the Dockerized Flask backends to seamlessly communicate with the host-level shared MCP and RAG servers without breaking existing network topologies. Crucially, Release 0 functionality remains completely intact: basic CRUD operations (e.g., opening/closing positions) and local AI-Mode risk analyses function as expected. The new RAG and MCP endpoints return strictly formatted JSON (including confidence catageories and source arrays for RAG, and structured tool outputs for MCP), which the frontends parse and inject directly into the DOM using HTMX, avoiding full-page reloads and maintaining a fluid user experience.

7. Individual Contributions

Student	Release 1 Work Completed	Integratation/Validation Activities	GitHub Commits
Student 1	CODE: ❑ Built the shared live data layer for Polymarket and Manifold (Later raplaced Manifold with Kalshi by Student 4) with caching. ❑ Developed the shared MCP server and registered MCP tools.	- Tested the shared MCP server with live market data. - Tested the shared RAG server with grounded	6

	❑ Developed the shared RAG server with retrieval, citations, and confidence scoring. ❑ Integrated MCP and RAG into the Student 1 frontend → backend flow. ❑ Extended the agentic loop with MCP/RAG validation. ❑ Added MCP/RAG environment variables to the Docker Compose services. ❑ Updated CI configuration to retain MCP/RAG code while disabling those steps in CI. ❑ Validated R0 CRUD and AI-Mode functionality after Release 1 integration. Individual Report Sections: • Created R1 Report Template including report structure • Created a Todo list for both Code and Report section, separating individual task and shared claimable tasks to evenly distribute work. Shared Report Sections: • (SECTION 1.3): Feature Updates and Responsibilities • (SECTION 6.2): Per-Feature Validation Evidence	questions, citations and confidence. - Tested the agentic loop with MCP/RAG validation. - Tested MCP/RAG integration through the Student 1 frontend and terminal. - Confirmed R0 CRUD and AI-Mode still work. - Tested Docker Compose and verified MCP/RAG environment variables.	
Student 2	CODE: • Added MCP and RAG access to the Portfolio & Position Tracker. New backend routes POST /mcp/execute (read-only tool allowlist: validate_portfolio, get_exposure_summary) and POST /rag/ask (query validation, max 500 characters). The frontend has "Portfolio Validation" and "Market Intelligence" panels that show the structured tool result, and the RAG answer with citations, a confidence label and an insufficient-context message. • Shared: added two MCP tools to the shared MCP server.	Integration/Validation Activities • 14 automated tests pass. They cover CRUD, price history, the MCP/RAG disabled paths, the tool boundary (unknown tool rejected) and empty-query rejection. They also cover the risk-analysis Observe/Adapt paths: removing a disclaimer, retrying after a refusal, and the calculated fallback. • Tested the MCP and RAG servers directly in the terminal (/mcp/call, /rag/query, /health) and	5

	validate_portfolio checks prices, sizes and sides, flags duplicate and self-hedged positions, warns when one position is more than 50% of a portfolio's cost, and returns cost basis, max payout, max loss and max profit. get_exposure_summary returns exposure grouped by market category. • Shared: added load_portfolio_entries() to the shared RAG server. It puts positions, 30-day price history, trades, an exposure summary and the newest position into the knowledge base used by all four features. • Added a price_history table with seeded data, and a GET /positions/<id>/history endpoint behind a Chart.js P&L and price chart. • Extended the Release 0 risk analysis agentic loop. Observe now detects LLM disclaimers and checks the answer mentions real tickers. Adapt strips the disclaimer, retries with a stricter prompt, or falls back to a summary calculated in code. • Added backend input validation for new positions (HTTP 400 on bad price, size or side). LLM and database output is now HTML-escaped (XSS fix). The portfolio dropdown is now filled in server-side.	through the Student 2 backend routes. • Tested end to end in the frontend on the running Compose stack: • Validation: VALID on Crypto YOLO with the 52% BTC concentration warning, and INVALID caught on a bad test trade. • RAG: a grounded answer citing portfolio:position#1, and an insufficient-context answer. • Confirmed Release 0 CRUD, the chart and AI-Mode risk analysis still work, including the agentic trace. • 12 successful student-2-ci runs.	
Student 3	• Added MCP serach access (POST/mcp/search) and RAG grounded Q&A access (POST/rag/ask) to the Market Research Notes and News feed backend/API and frontend UI, each implementing the full PLAN>ACT>OBSERVE>ADAPT trace and a graceful fallback if the	• Ran the full containerised stack locally via docker-compose up, performed MCP serach and RAG Q&A through the frontend UI (both a grounded and insufficient context	7e087cfd5192a2{PRs

	shared server is unreachable. Fixed a displaygap where the frontend did not distinguish an insufficient-context RAG response from a normal grounded one. Contributed load-notes-entries() to the shared RAG server so research notes and news articles are included in the RAG knowledge base used by all four features. Updates student-3.yml to keep MCP/RAG integrated but disabled (unreachable localhost URLs) during CI, added two new pytest cases that assess graceful fallback behaviour. Updated docker-compose.yml's student-3-backend service with MCP_URL /RAG_URL and host.docker.internal configuration	case) validated both routes directly with curl and ran pytest locally (6/6 tests passing including the two new MCP/RAG CI fallback tests) before pushing to the repository.	
Student 4	CODE: - Built cross-exchange mispricing for the AI Market Analyst Assistant: MCP search and compare, save as analysis, and "Explain this gap" through RAG. - Added backend routes POST /mcp/execute (read-only tool allowlist), POST /mispricing/save and POST /rag/ask (validated market_ids, insufficient-context handling). Each returns a Plan → Act → Observe → Adapt trace and a graceful fallback when a shared server is unreachable. - Added the "Ask with sources" panel with citations, a confidence label and an insufficient-context message. - Fixed an XSS bug: all database and service output is now escaped. - Shared: replaced Manifold with Kalshi in the live data layer (full background index of open events, month-name matching, bid/ask midpoint pricing) and fixed the Polymarket search. - Shared: added the compare_markets MCP tool (mispriced / watch / fair, edge after spread, last-trade fallback).	43 automated tests passing: 20 feature, 8 compare_markets, 15 data layer. The feature tests cover the disabled paths, the tool boundary, input validation and the live paths against faked servers. - Tested the MCP and RAG servers directly in the terminal (/mcp/call, /rag/query) and through the Student 4 backend routes. - Tested end to end in the frontend with live Polymarket and Kalshi pairs, covering both grounded and insufficient-context answers. - Confirmed Release 0 CRUD and AI-Mode still work, both in CI's container smoke test and locally. - Recorded the showcase demo.	9 commits, 3 PRs

	• Shared: added multi-market (market_ids) retrieval to the RAG server. • Shared: UI refresh across all four frontends (PR #16). • CI: student-4.yml with MCP/RAG disabled, plus offline data-layer and compare_markets tests and a container smoke test. Docker Compose: MCP_URL/RAG_URL via host.docker.internal. Report sections: 1.3, 6.2, 6.6 and 7 (Student 4).		
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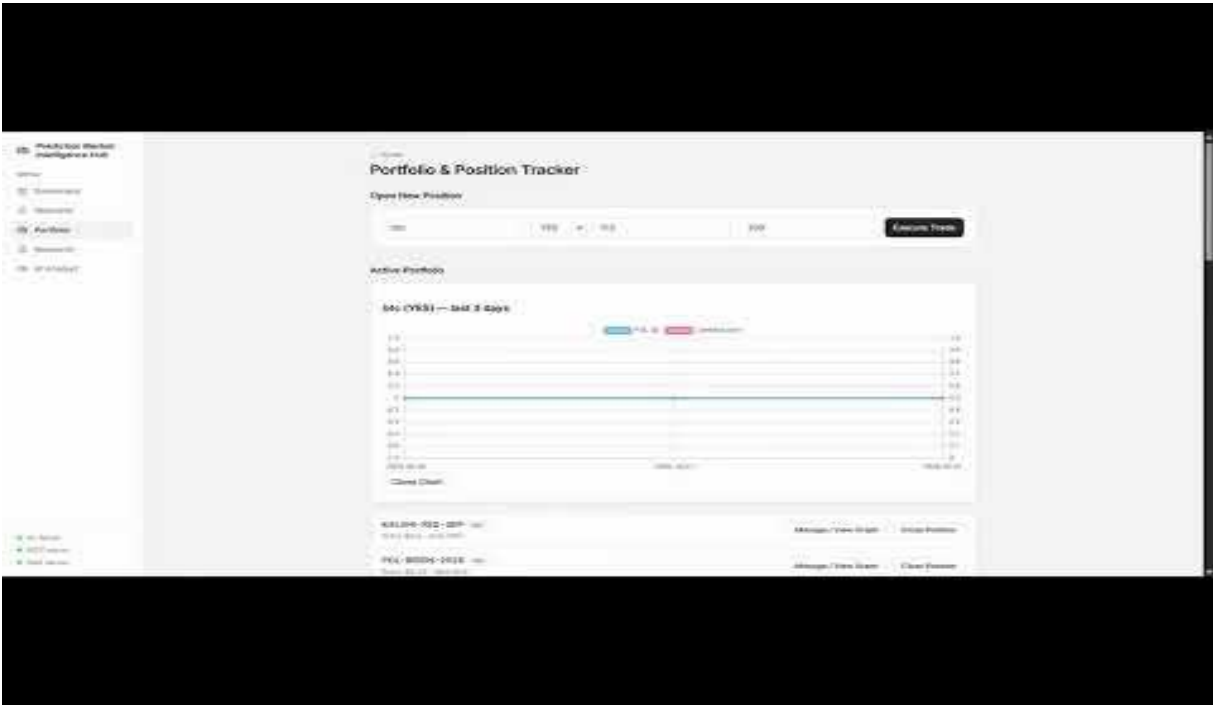
8. Repository and Showcase Links

GitHub Repository:

<https://github.com/Erfanur1/ADVSD>

Showcase Video URL:

<https://youtu.be/psCCvjfd-iU>



9. Known Issues and Limitations

- Carried over from Release 0: the database layer is embedded within each backend/API container rather than implemented as a fully separate database-API service per the original specification's three-tier description.
- The application now depends on the Polymarket and Manifold public APIs for live market data. A local cache is used to reduce the impact of API downtime or rate limiting, but a prolonged outage on either platform would still degrade the freshness of data shown to users.
- Kalshi is not used as a live dni
- Data source; only Polymarket and Manifold are integrated, since Kalshi requires account registration and is a regulated exchange.
- Manifold's data is play-money and community-generated, so its market quality/coverage differs from Polymarket's real-money markets; this distinction should be made clear to users where both sources are shown together.
- Historical Price Graphing (Student 2): The current iteration of the Chart.js profit/loss graph utilizes mock data arrays for the MVP. Full dynamic charting requires the backend SQLite database to be updated to store and serve minute-by-minute historical price action, which is slated for a future release.
- CI/CD Blindspots: Because the shared MCP and RAG servers run locally on the host machine and are not containerized, GitHub Actions cannot execute end-to-end tests for these specific features. CI workflows currently bypass RAG/MCP validation steps to prevent false-negative build failures.